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COLD FUSION TIMES

FUSION FROIDE : Les Prs Pons Et Fleischmann Ont Bien Trouvé Une Nouvelle Source D'Énergie

ICCF-5 PRESENTS > 130 PAPERS

Monaco -

The Fifth International Conference on Cold Fusion (ICCF-5) is now history. Over 200 serious experimenters and theoreticians assembled in Monte Carlo for ICCF-5 as they braved jet-set accommodation rates, exquisite beaches, and Baywatch-class scenery -- all in an attempt to concentrate on the April meetings on cold fusion sponsored by **IMRA EUROPE S.A.** [Cedex, FRANCE].

World-class scientists got together to learn more but also to work and share data. Several reports of the meeting are herein with a spectrum of opinions and a review of the reviewers!

"P-C" CF PATENT ISSUES

Washington D.C., Cambridge, MA. -

U.S. Patent number 5,411,654 has issued entitled "Method of Maximizing Anharmonic Oscillations in Deuterated Alloys" [filed July 2, 1993] and based upon the latent heat and anharmonicity theories of Dr. Keith Johnson [reported here over one year ago]. It is particularly ironic that the patent goes to **Massachusetts Institute of Technology** where some steadfastly denied the existence of cold fusion for over six years. Paradoxically the patent lawyers painted the patent application with a "politically correct brush" by soundly omitting the words "cold fusion" [although Brian Ahern, Keith Johnson and Townsend do cite F&P in their patent]. §§

ENERGY FUNDING GUTTED

FIRST SHIFT IS FROM HOT FUSION

Washington D.C. and New York -

As first reported in The New York Times [May 22, 1995; "GOP. Budget Cuts Would Fall Hard On Civilian Science"] science funding is taking a downward turn. The Federal Science establishment is being gutted under the proposed Republican budget from its current funding of $17.5 billion. The Republicans have proposed cutting that by $7 billion over five years.

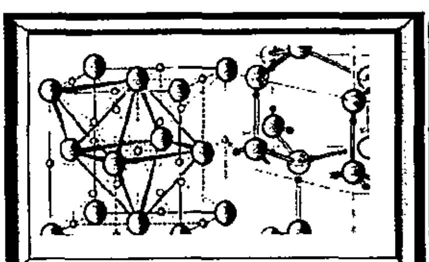
With most government subsidies presently going to big industry and the military, this shift has implications both for the security of the United States and the field of cold fusion. In addition to skirmishes for military bases, the Committee recommendation for Basic Energy Sciences is $792,661,000, a decrease of $18,758,000 [or >2.3%] from the budget request of $811,419,000. [FY95 appropriation: $733.9 million.]

The most exciting news: $1,000,000 to be provided to fund research on the potential energy applications of sonoluminescence, an effect in which highly concentrated sound waves in liquids generate very short bursts of light from cavitating bubbles.

MUCH PROGRESS IN COLD FUSION

Utah and Massachusetts -

Since the announcement of the discovery of cold fusion (CF) in March 1989, laboratories in over 30 countries have confirmed the CF phenomena. This investigative effort has resulted in a voluminous number of cold fusion papers; many published in peer-reviewed journals, presented at technical conferences or published in technical periodicals. (Cont. page 4)



Palladium must be fully filled with deuterons to generate the reactions. The face-centered cubic fully loaded metallic structure is shown on the left, totally filled with deuterons obtained from the heavy water on the right hand side.

O.J. Simpson and Cold Fusion page 6

TOWARD THE FUTURE

Dr. Richard Oriani Addresses IT Alumni

"*Take CF Seriously*", Advises Chemist

[The review of this address was printed in the Univ. of Minnesota's Office of Technology Transfer's "Research Review." Vol. 24, No. 8, Feb. 1995 [Minneapolis, MN]. We are grateful to Aaron Patula in reporting and posting it on the Internet.]

Minnesota- Cold fusion, Richard Oriani told an audience of Institute of Technology alumni, "is certainly worthy of study and funding." There are good reasons for skepticism, he acknowledged, but there are also good reasons for genuine interest. "Here is some new kind of nuclear physics, and it is too late to heap ridicule on it," he said. Oriani, professor emeritus in the University's Department of Chemical Engineering and Materials Science, spoke at a December 7 seminar sponsored by the **IT Alumni Society**. He wished to give his audience "an appreciation of where cold fusion research is after these five years," he said. "People have made a lot of headway."

When nuclear reactions release energy, Oriani's explanation of the data began, it is because some part of the mass involved is converted to energy. For example, in one of the reactions theoretically associated with cold fusion, an atom of deuterium combines with an atom of tritium to yield helium, a free neutron, a decrease in mass of 0.0188 atomic mass units (AMU), and energy at the rate of 1.49×10^{-10} joules per AMU (equivalent to 8.97×10^{13} joules per mole).

In the five most accurate energy- measurement experiments, the energy output ranged from 106 percent to 170 percent of the energy put into a palladium- deuterium system. For two groups of experimenters, there was a net gain of energy every time they tried the experiment. Oriani's group produced a net gain in two attempts, but thirty subsequent attempts produced no energy. The inconsistent results, said Oriani, seem to depend on the sample of palladium. His third success came after the thirty failures when he obtained a new sample of palladium from a Japanese source. Other groups have measured, in three less accurate experiments, energy production ranging from 5 to 15 times the energy input.

When Oriani tried to publish his own experiments, he said, the two journals' replies were to the effect of "We already know cold fusion doesn't work, and you don't understand your results. We're not going to publish them."

Oriani's reply: "Many things are published without full understanding, and that's the way it should be."

Oriani's second set of reports dealt with observed effects that could only result from nuclear reactions. For example: Fritz Will et. al., electrolyzed heavy water with cathodes made of palladium from two different suppliers. (Heavy water is D_2O, i.e. water containing deuterium rather than common hydrogen. Will was director of the Utah Cold Fusion Institute). One type of palladium yielded no tritium. The other type yielded tritium at 50 times the background level, in four trials out of four. From that second type of palladium, 140 samples not subjected to electrolysis were found to contain no tritium.

Melvin Miles and Benjamin Bush, using palladium and heavy water, produced helium in concentrations ranging from 5.4 parts per million to 9.7 ppm. The background concentration of helium in air is 5.2 ppm. George and Stringham, using sound to cavitate heavy water on palladium foil, produced helium at 10 times background levels in ten trials out of ten. Y. R. Kucherov, et. al., by means of "glow discharge" with a palladium electrode in low- pressure deuterium gas, produced helium at 4 to 100 times background levels and counted 10 raised to the 7 neutrons per second.

Skepticism and ridicule of cold fusion began in 1989, Oriani remembered, when Stanley Pons and Martin Fleischmann announced their discovery through publicity rather than peer review. "They described their work so poorly it seems they wanted to keep it obscure," said Oriani. His own interest in cold fusion was sparked shortly after that, by the work of Steven Jones at Brigham Young University. Since the Pons and Fleischmann debacle, cold fusion experiments have not been adequately published, Oriani argued, because the journals Science and Nature have been "caustic and abusive" toward the work.

§§

COLD FUSION RADAR

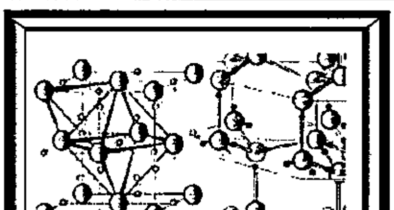
In this issue we salute Fusion Facts and "Cold Fusion" for making it another year. Infinite Energy is to be congratulated for the best improvement in pictures in a cold fusion periodical. **There are at least four (4) cold fusion periodicals now supporting the periodic reports and commentaries in Fusion Technology and other popular magazines.** ICCF-5 was important and another boost in the development of this field. In this issue are comparative reviews from Jed Rothwell, Bill Paige, Bill Bass. We have reviewed them all, and selected excerpts of the best. Readers are encouraged to read the full papers in the respective magazines, and to consider the evidence to date.

Best factoid
Dr. Edwin (Buzz) Aldrin took the time be a Guest Observer at the meeting.
[*Infinite Energy*; no attribution]

The COLD FUSION TIMES' BEST of the ICCF-5

BEST ABSTRACTS -- (tied) FUSION FACTS and "COLD FUSION"

BEST PICTURES ICCF-5 -- Dr. Eugene Mallove
BEST WRITE-UP OF MEETING -- Jed Rothwell



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POWER PROFILE OF THE MONTH

Prof. Robert Bass (California)

-- He had the privilege of hearing the definition of a **TRUE** scientist from the lips of Albert Einstein himself. This occurred at a 1950 sailing party at the home of Frank Aydelotte (Director of the IAS) for the Rhodes Scholars-elect when he was 20 years old. The Nobel Prize winner said, "If I could give the young men any advice, it would be: Never believe anything just because you hear it on the radio or see it in the newspapers or everybody else believes it! **ALWAYS THINK FOR YOURSELF!!!**"

Dr. Bass has reminded us that today this is excellent advice for one who wants to be a "true" scientist, but possibly devastatingly awful advice for anyone who is just a politician with some scientific training, that is for one "who is merely career-minded, including academic freeloaders". Subsequently, Dr. Bass served three years of postdoctoral study at Princeton under a National Medal of Science winner, Solomon Lefschetz. While there he worked on nonlinear mechanics, topological dynamics including structural [robust] stability in dynamical systems. He drifted towards increased rates of innovation, and by 1974 his Topolotron patent application (issued in 1985) contained teachings of what would eventually become several of the biggest discoveries of fusion plasma physics during the 1985-1995 decade. §§

ICCF-5 and the CF SKEPTICS

Dr. Robert Bass

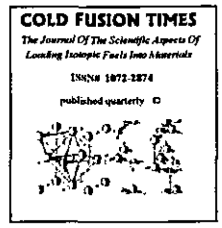
I just got back from ICCF-5 at Monte Carlo. The James Patterson & Dennis Cravens cold fusion cell (using ordinary water and nickel cathodes) was producing 3 times more energy out than in during the 4 days of the conference. It starts up in about 10 minutes and is reliable. Patterson has got 3 issued patents because in his PTO applications he never mentioned 'cold fusion' and finessed the question of energy generation. Otherwise the PTO would have sat on it as they have [unconstitutionally!] sat on 200 other cold fusion patent applications for the past 6 years.

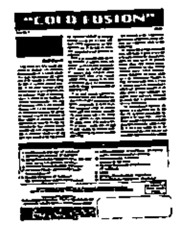
Imagine the subject you love and value most, and then think of an **arch-enemy** of that subject, who will do **anything** to suppress it (and in this example will be bad-mouthing it until the day that Cold Fusion-Powered cars are being sold by Toyota!), and then imagine him being able to spread his propaganda freely through the disinformation machine of the Establishment! The skeptic will claim that cold fusion is impossible' because when deuterons are as close as they are in deuterium molecules (or liquid deuterium) they do not fuse. This is utter incompetence. He is neglecting the **periodicity** of the deuteron lattice inside a host metallic lattice (such as palladium or titanium). The phenomenon of Resonant Transparency of Coulomb Barriers is well understand in Quantum Mechanics (see Bohm's classic QM book) and is well-established experimentally in both Atomic Physics and Nuclear Physics. Skeptics treat the problem as if it were a purely **local** instead of a **global** problem and have ignored the main theorem of Solid State Physics (Bloch's Theorem) which says that a solution of Schroedinger's equation inside a periodic lattice is irrelevant unless its logarithmic derivative is **periodic** in space with the same period as the lattice! Morrison [ed. CERN's {the European hot fusion group's chief skeptic against cold fusion] also ignores the massive evidence that d+d goes to helium-4 inside palladium lattices. Enough He-4 to account for the heat has been measured repeatedly by Mel Miles at CLNWPS [Ed. China Lake Naval Warfare Lab] and by Gozzi at the U of Rome! This measurement is difficult, because the amount of helium that will be produced is less than the ambient trace-helium near many chemistry labs; stainless steel vessels must be used, etc.

Morrison also ignores that hundreds of researchers (mainly in Japan, Italy, India and some independent mavericks in the USA) have reproduced the F&P experiments and that there have been peer-reviewed papers in archival journals by internationally recognized calorimetrists pointing out the numerous errors made in the "Big 3" negative experiments (MIT, Caltech, and Harwell). The reanalysis of Harwell's results by others (to whom Harwell loaned their equipment) is gradually forcing them to the edge of admitting that they analyzed their results mistakenly, and they can be expected to retract their negative report soon. §§ (More on page 8)

OF GREAT INTEREST TO OUR READERS

SEE FOR THE WORLD (PAGE 8)





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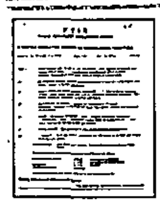

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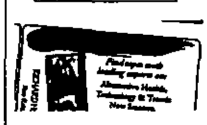

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PO Box 3010
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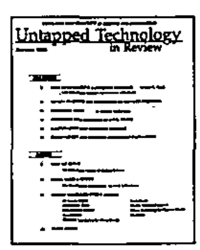
98009-3010



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CATALYTIC EFFECTS IN COLD FUSION

Peter Glück
Institute of Isotopic and Molecular Technology

Cluj-Napoca, Romania - The emergent new class of phenomena known as cold fusion is missing a definitive, complete, quantitative theory or even an explanation, like other neighboring fields of solid state science as : high temperature superconductivity, conductive polymers, heterogeneous catalysis, porous silicone. All these are developing mainly by steady accumulation of know-how. A distinctive feature of cold fusion is the difficult reproduction of the experimental results in all the systems developed till now: electrochemical, ultrasonic, gas/solid. Based on a breadth-first approach and on positive thinking, the problems of reproducibility are regarded as an essential information/message:cold fusion is an extreme form of catalysis, plagued by hypersensitivity. However, this is not the karma of CF because perfect reliability can be achieved by adequate, rational, technical solutions. Today the dominant CF paradigm, originating in hot fusion is: "Everything happens inside the lattice, and only in the lattice, in the whole lattice. Energy is transferred to the lattice." The alternative paradigm, based on catalysis is: "Everything happens on the surface, and/or very near to it, and only in certain active sites. Energy is transferred to the surface electrons." [1].

The entire body of published data contains evidence for the surface character of the CF processes (efficiency of high surface cathodes, direct, decisive influence of some impurities, expelling of the reaction products, non-specific effects), for the existence of restricted active sites (bursts, in vivo and post-mortem investigations) as well as for the dynamic stimuli for triggering and quenching of the processes (parameter variation ,cavitation, pulsing, electromagnetic effects). Recent remarkable catalytic effects in cold fusion are addressed as the "heat after death" effect [2], the impressive effects of the nanometric materials with huge [3] and protected [4] surfaces and direct proof of the existence of restricted active areas [5]. Some positive, excess heat experiments have been carried out in our Institute.

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MUCH PROGRESS IN COLD FUSION

Hal Fox, Fusion Information Center Mitchell Swartz, JET Technology

Massachusetts, Utah -- (continued from page 1)

Most importantly, there is a cornucopia of new methods by which nuclear reactions can be controlled in relatively novel devices. Although the first reports cited the use of heavy-water electrolysis with palladium cathodes and lithium salts, these methods now include light-water electrolysis with nickel cathodes and alkali-metal carbonate and other solutions, molten salt electrolysis with palladium as the anode, gas-plasma discharge devices incorporating palladium target cathodes, devices using a variety of proton conductors, hydrogen gas loaded into heated nickel reactors, and several acoustic techniques including collapsing bubbles vicinal to palladium metal into which the hydrogen is injected. Yet, in spite of the rich phenomena now replicated in a diverse number of ways, there has remained an unprecedented reluctance on the part of many peer-reviewed journals to publish cold fusion articles.

Fortunately, the time-rate of change of peer-reviewed publications in the field (often preselected so as to only include the peer-reviewed papers and those which initially were unable to reproduce the requisite difficult-to-attain conditions required for CNF) appears to be increasing[4] despite the obstructions to dissemination of the scientific information. That data in now augmented by the formidable data base of more than 2000 papers[5] collected by the **Fusion Information Center**, which may be a complete collection of the world's literature on CNF. Further supplemental information includes the corporate entities in this developing field. To bypass the reluctance for reasonable publication in conventional journals, to hasten the spread in the awareness of these new scientific theoretical and experimental results, we have suggested plans for additional on-line computer-accessible information to supplement the existing journals and newsletters in this field.

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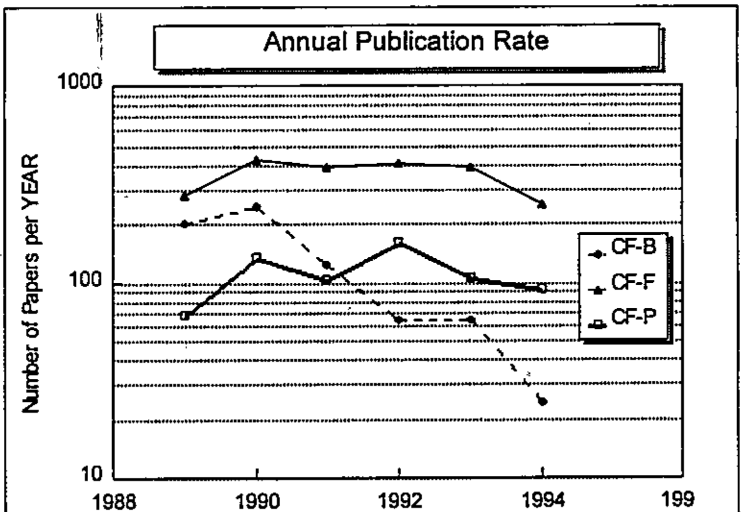

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Three DATA Sets of Published Data in Cold Fusion

The cumulative CF literature and the publication rate of the papers is shown as a function of time from the initial announcement. The Dieter Britz selections are labeled CNF-B. The formidable database (CNF-F) refers to the more than 2000 scientific papers collected by the **Fusion Information Center**. From these 2,142 papers, there are positive results (success) in 664 papers (labeled CNF-P).

CF-F and CF-P demonstrate achievement in replicating and advancing the cold fusion technology.

COLD FUSION THEORIES

SOLVING THE PUZZLE OF EXCESS HEAT WITHOUT STRONG NUCLEAR RADIATION

Li, Xing Zhong Department of Physics, Tsinghua University

Beijing, CHINA -- After 5 years of intensive investigation, the excess heat from the deuterium in condensed matter under ambient conditions has been reproduced in various schemes (open system, closed system, and closed system with no recombination). At the same time, the nuclear measurements showed that there is no commensurate neutrons, and no g-radiation; the helium and the triton signals appear in an unconventional branch ratio. Does this anomalous phenomenon conflict with our 50 year knowledge of the nuclear physics? The Gamow penetration factor of the Coulomb barrier is the first problem, and the branch ratio after the penetration of the Coulomb barrier is the second problem. The Gamow factor was first introduced for the case of the free-particle injection, or for the case of the free-particle emission. If the reactant particles are both confined in the solid, and if the energy level in the lattice potential coincides with the energy level in the nuclear interaction well; then, the penetration factor should be q^{-1} instead of the traditional Gamow factor q^{-2} (q is a large number determined by the thickness and the height of the Coulomb barrier.) Usually, the idea of resonance penetration has been considered not applicable, because the difference between the confined particle and the free-moving particle has not been paid enough attention to. For the free-moving particle in resonance, the only requirement is that the energy of the incident particle has to match with the virtual energy level in the nuclear well. The absorption in the nuclear well might be arbitrarily strong. However, in the case of a confined particle in resonance, not only the energy level in the lattice well has to coincide with the energy level in the nuclear well, but also the imaginary parts of the potential well are limited by the boundary conditions. In reality, the absorption in the nuclear well has to match with the penetrating flow from the source in the lattice well, in order to keep the conservation of the current of the probability. If the absorption is too strong (stronger than q^{-1}); then, there will be no resonance penetration. Just due to this reason, the fast nuclear reaction channels (e.g. d+d --> ^{3}He +n) would not be in resonance, and the slow reaction channel (e.g. d+d --> ^{4}He + heat) might be in resonance. Since the resonance would enhance the penetration rate by a factor of q, what we see in the deuterated solid system is the slow reaction channel; i.e. excess heat and helium without the commensurate nuclear radiation. §§

ENERGY GENERATION MECHANISM IN UNITARY QUANTUM THEORY

Lev G. Sapogin Department of Physic
Technical University (MADI)

Moscow, Russia -- It is now a well established fact that in Cold Nuclear Fusion (CNF) only a small portion of heat results from nuclear reactions, the rest being of a mysterious origin. In this connection Prof. Peter Hagelstein writes in [1]: "Some say that this heat can be explained easily by elementary chemical reactions, phase changes, or battery-like storage effects. I have trouble with these explanations". For instance, nickel electrolysis in lightwater produces the same amount of energy as that of palladium in heavy water. Besides, we have to consider a no less mysterious phenomenon of sonoluminescence, that was discovered in Russia in 1933 by S.N.Rzhevkin. At first sight these phenomena seem to bear no correlation. But Julian Schwinger, the Nobel Laureate and profound research worker, has drawn parallels between cold fusion and sonoluminescence in his continuous technical publication on both topics. He notes in [2]: "Like Cold Fusion, sonoluminescence "should not exist", but it does. This now well established phenomenon occurs when ultrasonic sound, beamed into liquid, causes bubbles to oscillate stably - to expand and contract regularly -and also to emit regular pulses of light". But one is positively struck by the heat generator built by Dr.Yu.S.Potapov (Kishinev, Moldavia), which produces in water cavitation bubbles, generating 12 kW of heat on 4 kW of electric energy. This is achieved by the aid of a centrifugal pump that runs light water thought a tube with a number of special attachments. Similar, though less effective results, have been obtained in the USA by James Griggs [3]. The reactions taking place in the above-mentioned cases can be explained neither in chemical nor in nuclear terms. Moreover, Potapov's system is now in production and used for heating homes. There is no doubt that the described phenomena can be viewed as "new physics" and it is impossible to describe them within the laws of conventional science.

Below it will be shown that Unitary Quantum Theory (UQT) allows one to explain all these seemingly unrelated effects. UQT was developed in 1970-1988 and its validity is proved by the fact that at a limited number of cases the theory results in both the Dirac's equations and the relativistic equation of Hamilton-Jacobi. Moreover, the solution of approximate UQT equation has yielded the electric charge value and that of the fine structure constant with reasonable exactness, the result being achieved for the first time in theoretical physics [6-14]. The solution allowed to formulate one more approximate equation for the oscillating charge particles model which gives promising results in dealing with the deuterons interaction problem (the basic CNF problem) [4,15].

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THE PATENT PAGE

Analysis of the Intellectual Property Field in Cold Fusion -- see page 1

Highlights of the Fifth International Conference on Cold Fusion (ICCF-5)

Jed Rothwell -- COLD FUSION TIMES WINNER : BEST ICCF-5 REPORT
[Mr. Rothwell has served the United States as he has translated and reported on world-wide events in the field of cold fusion.]

Monaco - Wide-ranging positive results in both excess heat and nuclear products were reported from E-Quest, U. Milan, Osaka National U., Mitsubishi Heavy Industries, NTT, the Japanese National Laboratory for High Energy Physics (KEK), Los Alamos, BARC, Amoco Production Company, Shell Oil, Harwell and others. An electrical engineer from Bechtel Corporation gave a superb talk on the economic and technical aspects of the commercial development of cold fusion energy. A live demonstration system from Clean Energy Technologies Inc. showed 300% to 1000% excess energy. This is a brief report of my impressions of the **Fifth International Conference on Cold Fusion** (ICCF5), April 9-13, 1995, Monte-Carlo, Monaco. First, a word about media: the Book Of Abstracts is about 150 pages. It is well organized for once. Abstracts are numbered in a coherent scheme. The first lecture was the Critical Overview by Storms [1]. It was one of the best. Patterson's company, Clean Energy Technology (CETI), got together with Dennis Cravens and brought to the conference a demonstration cell in a flow calorimeter. It worked spectacularly well. Cravens [2] discussed it on the first day. The device output 3 to 5 times input energy, ignoring energy lost to electrolysis gases, and as much as 10 times input if you include various factors like electrolysis gases and the heat lost from the cell container.

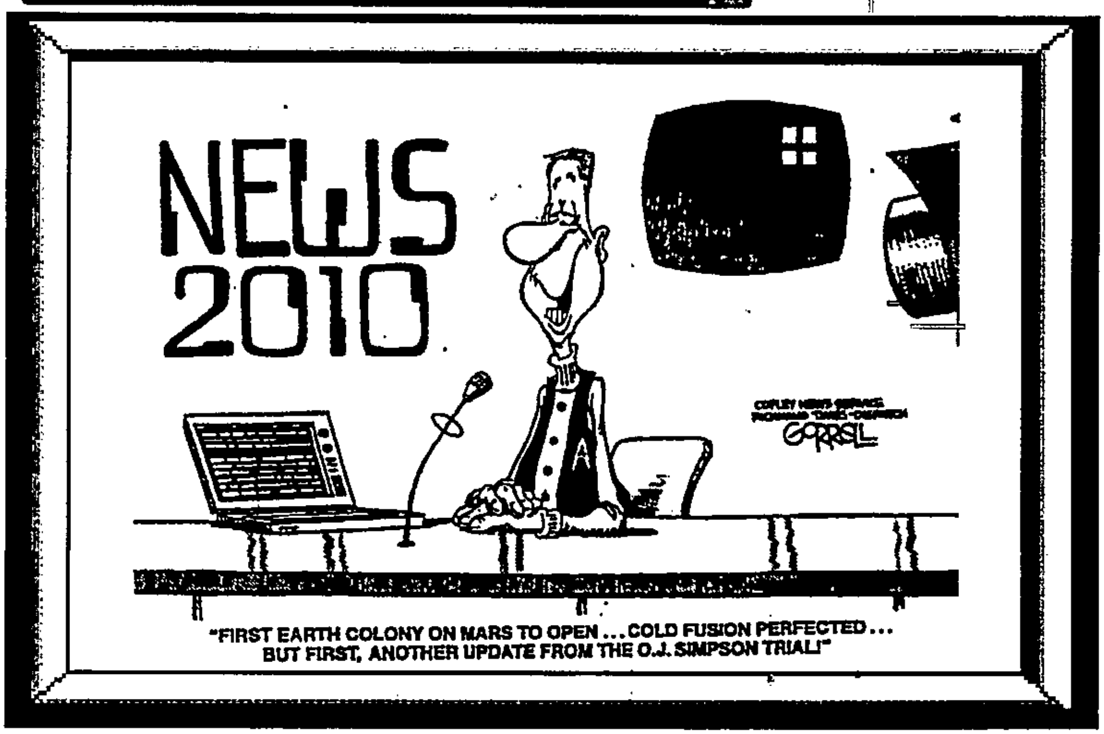
Unlike the majority of the skeptics, Mr. Jed Rothwell examined the evidence

Briefly, input was usually held at about 0.4 watts I*V, although on the last day it was raised to 0.8 watts for a while. The flow rate was 10 ml per minute. When the machine was first rolled into position and turned on in the morning, there was no excess for 10 or 20 minutes, and the temperature Delta T fluctuated around 0.2 deg C, indicating about 0.14 watts output. The rest was lost to known heat leaks from the cell container and to the effluent gasses from electrolysis, which were measured in a gas flowmeter. As the reaction turned on, the Delta T gradually rose to about 2 deg C, and sometimes rose as high as 4 deg C, indicating 20 to 40 calories per minute, or 1.4 to 2.8 watts.

Patterson's device is described in U.S. Patents 5,036,031 and 4,943,355. It is a thin film light water system. It incorporates plastic beads coated with Cu, Ni, Pd, and another layer of National Laboratories and Universities. A few months ago they had problems with high temperatures and pressures destroying the beads and melting the plastic cell containers. They designed new beads with an extra outer layer of nickel, and they found new, temperature resistant cell materials.

The CETI demo system is fairly predictable, well controlled, and well behaved, although it did get a bit quirky in the harsh conditions of the ICCF5 hallway. During breaks, the hotel coffee pots kept tripping the circuit breakers. This sent jolts of power through the transformer, which crashed the experiment. The CF reaction started up again every time, usually in about 10 minutes. The high precision flowmeter unfortunately did not survive the beating, the batteries and power supplies in it burned up. Fortunately, the low precision flowmeter -- a 10 ml lab supply graduated glass cylinder plus stopwatch -- cannot be affected by power outages and excess voltage. The experiment was subjected to other abuses: the cart holding the experiment was wheeled up to a hotel room every night, carried on elevators, and pushed around. Cravens even lifted the cell from its container to show it to people while it was running! Yet in spite of this, the reaction started up in the morning after 10 or 20 minutes of electrolysis, although on the last day it took about a half hour, and the power was turned up higher than before. The fact that the cell survived this treatment at all demonstrates that this is one of the most robust and practical electrochemical CF systems yet developed. By the last day, the batteries in the differential electronic thermometer got weak and a minor 0.2 deg C bias appeared between them, which could be observed by switching the leads to the input and output thermistors. This was not significant, because the Delta T ranged from 2 to 4 deg C when the reaction was on, and it was less than 0.4 deg C when there was no excess heat. (continued on page 9)

OUR FAVORITE CARTOON OF THE "YEAR"



ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

Why was there initial interest in cold fusion but not later?

Answer: In addition to energy, tritium may have had a role.

"Mr. Frank Close, in his 'Too Hot To Handle' book of 1991, put out by the Princeton University Press, has an interesting section on pages 49-51 about the tritium stockpile of the United States. It seems the U.S. has on hand about 100 kilograms of Tritium stockpiled, including what is in the fusion bombs (missiles like Poseidon -- etc.) And 5-6 kilograms of this Tritium has to be replenished each ear due to the half-life of 12.3 years. If nothing is done to replenish, you have automatic nuclear disarmament going on. The existing nuclear reactors operated by the DOE that supplied the Tritium were getting old (one of them: Savannah River Reactor) and being shut down for repairs. By 1989, the search for tritium was becoming critical (militarily). Then when the announcement for cold fusion came out, DOE (and the military) was looking for the tritium production possibility rather than the energy production aspect ballyhooed in the press. And since the cold fusion effect was aneutronic, dropped it like a 'bomb'. They had other priorities in mind."

[aki@ix.netcom.com (Akira Kawasaki) "Tritium, Cold Fusion, and the Military"25Feb1995,<3inkj9$g6f@ixnews3.ix.netcom.com>]

Evidence for Tritium Generation in Self-Heated Nickel Wires Subjected to Hydrogen Gas Absorption/Desorption

M. Srinivasin

T. Sankaranarayanan, M. Srinivasin, M.. Bajpai and D. Gupta
Bhabba Atomic Research Centre

Trombay, Bombay India -- A programme to study hydrogen gas loaded nickel samples was initiated following reports of observations of anomalous excess heat in such systems by Focardi et al. Nickel samples were in the form of electrically heated wires (0.125 mm or 0.38 mm dia x 500 mm length) coiled as a spring. The Ni spring was suspended inside a glass "cell" connected to a vacuum unit and gas handling system provided with a sensitive manometer for differential pressure measurements and could be heated electrically through tungsten leads. The lower part of the cell had an outer jacket through which coolant water could be passed for calorimetric measurements. For outgassing/deloading H_2 gas, the wire was raised to glow hot conditions under a vacuum of 10^{-5} cm of Hg, while for loading, Iolar grade H_2 gas at subatmospheric pressure was introduced into the cell. After several trial runs, an appropriate protocol for activating the Ni surface and obtaining a maximum rate of absorption was established.

As recommended in the literature, repeated cycles of loading/partial unloading were carried out to increase the net absorption of H_2 into the wire. A maximum net loading of over 0.3% by weight of the nickel wire could be obtained after about a week of pressure/temperature cycling. At the end of the loading runs, the wires were cut into 3 or 4 pieces and dissolved separately in 5 ml of dilute HNO_3. After neutralisation of excess acid, the solutions were vacuum distilled prior to liquid scintillation counting for determination of tritium content. For this, 1 ml of the distilled sample was added to the scintillation cocktail. So far, 6 out of 9 loaded wires have indicated generation of tritium. However, not all the cut pieces from a given wire indicated presence of tritium, suggesting that tritium production is non-uniform over the length of the wire. For example, out of 27 cut pieces which have been dissolved and counted up to now, only 8 pieces have shown tritium in the range of 1 to 2,700 Bq. For the lowest activity case of 1 Bq/5 ml, the count rate was approx. 10% above the background value of approx. 250 counts/10 min.

The maximum amount of tritium production, namely 2,700 Bq, was observed in one 11 cm segment of a 45 cm long wire which displayed exceptional absorption/desorption characteristics in terms of both rate as well as quantity of H_2 absorption (>3% of Ni wire weight).

In case of one of the cut pieces of another sample wherein the surface layers were leached out separately prior to dissolving the balance portion, both the solutions indicated the presence of tritium. Blank (or control) samples cut from the two nickel stock spools dissolved and counted following an identical procedure have not given any counts above background levels. One cut piece from a loaded nickel wire when exposed sandwiched between two photographic films gave weak but identical spotty auto-radiographs, in both the upper and lower films. The present results thus corroborate production of tritium during electrolytic loading of natural hydrogen into nickel cathodes, first reported by us at ICCF-3 and confirmed subsequently at ICCF-4. Calorimetric measurements carried out to date during loading/unloading cycles with some of the wires have not revealed any anomalous heat generation in excess of V*I supplied to the nickel wires. Various techniques of "triggering" excess heat are presently being explored. § §

MATERIALS CORNER

This issue: TRITIUM PRODUCTION FROM COLD FUSION

In coming issues:
PROTON SEMICONDUCTORS -- MORE MATERIAL INFO
HELIUM MOVEMENTS THROUGH MATERIALS
ZERO POINT ENERGIES -- STANDARD ASH

COMMENTS ON FUSION'S FUTURE

by Robert Bass

The politicized ex-scientists who run the DOE fusion program have made the tragic mistake of ignoring two options which would make hot fusion power important for humanity in **certain** contexts. Their first mistake is to freeze on the Tokamak design, which Prof. Lawrence Lidsky characterized in the MIT-edited "*Technology Review*" as analogous to a "project to employee 2,000 Ph.D.'s for 40 years to design the world's first coal-fired, cast-iron biplane!". For this he was dismissed from his job as Associate Director of the **MIT Fusion Plasma Lab.** The intellectual bankruptcy of the DOE program is shown by the fact that they cannot stabilize the plasma long enough to go to "advanced fusion fuels." If they had really solved the stability problem they would not be so desperate as to employ 50% Tritium, and use a fuel cycle in which 80% of the energy is released in the form of a neutron flux, ensuring radioactivation of surroundings & embrittlement of the container, plus robot handling & a nightmare of other problems.

Some wiser heads like George Miley have advocated both "advanced" (aneutronic) fuel cycles (like boron & protons, or the correct mixture of helium-3 (He^3) & deuterium), which require higher temperatures and so a greater mastery of plasma stabilization technology. We have advocated 'remotely sited, pure deuterium-fueled, He^3 breeder reactors', namely near-unity-beta pure-deuterium-fueled, steady-state 130 keV **HOT** fusion reactors, in which only 20-30% of the energy released is in the form a neutron flux, and besides power, commercial quantities of He^3 are produced. The optimal temperature for such a reactor is 130 keV, at which temperature there are **two** equilibria (one thermally stable, the other thermally unstable!), and at which temperature the Lawson Number is less than half of what is stated in the literature!

The obvious proposal is to locate such a pure-deuterium-fueled hot fusion reactor in a walled, guarded, remote site in the desert (even though it is quantitatively demonstrable to be 10^5 times less hazardous than a fission reactor), to pacify the frightened public, and then export both energy and He^3; Miley et al have demonstrated that one such remote breeder can fuel **five**(!) equal-size [in Megawatts] Satellite Reactors in the middle of cities; but the metropolitan He^3/deuterium fueled hot fusion Satellite Reactors would be so non-hazardous that environmentalists would love them.

Cold Fusion Reactors will supply the bulk of mankind's future energy needs. In the F&P reaction (aneutronic $d + d$ to He^4) the great amount of the energy comes from the familiar Einstein relation $E = Mc^2$. [As Dr. Bass stated in 1992 at the DOE's Fusion Energy Advisory Committee meeting, this Nations'] "present course of ignoring cold fusion in the hopes that it will go away is **suicidal!** There are plenty of **legitimate** national projects (in national defense or in NASA's exploration of the solar system) wherein either the potential structural compactness of a hot fusion reactor or its possible high energy density (Megawatts per cubic centimeter) will offer engineering advantages with which cold fusion can never compete! §§

MORE HIGHLIGHTS ON ICCF-5

(continued from page 6 -- excerpts from
Jed Rothwell's **WINNING REVIEW** of ICCF-5)

Griggs [7] gave a surprisingly well received talk about his ultrasound device. His biggest improvement a dynamometer -- a "Lebow" brand Eaton torque sensor model 1805-5K. This gives him a second, independent method of measuring input. He has found that the mechanical power from the motor closely matches the manufacturer's performance specifications. Recently, after he modified the rotor, Griggs began experiencing problems with cavitation damage and with massive plating out of copper from an unidentified source. He has been working with experts from Georgia Tech and NASA to resolve these difficulties. They plan to use transducers, high speed cameras and other high tech tools to learn more about what is happening inside the machine.

Stringham and George of E-Quest [9] are getting massive helium, isotope shifts, heat and so on. Last summer they ran experiments at Los Alamos. E-Quest shipped samples of gas from the Los Alamos experiment in stainless steel collection bottles to Rockwell's facility in Canoga Park, CA. The Rockwell tests revealed definitive proof that the excess heat comes from a nuclear reaction. Experiments that did not generate excess heat showed 0.4 ppm helium. Experiments that did generate excess heat yielded helium far above that background level, at levels as high as 552 ppm, 100 times atmospheric concentration. Rockwell also looked at the ratio of ^3He to ^4He as well as ^{22}Ne to ^4He in the samples and found the isotopic ratios prove the helium could not possibly have come from contamination from normal terrestrial helium. During the talk, George mentioned that an extensive SIMS analysis performed by him at Lawrence Berkeley Labs showed dramatic isotope shifts in some of the high Z trace metals, especially titanium. It is one of the most important experiments in the field, so it deserves wider replication and exposure.

Arata [10] described his double structured cathode palladium black experiments in considerably more detail than his two most recent papers. He reported "the chemical reaction energy of 0.1 mole Pd-black used is only 4 kJ, but more than 200 MJ of excess energy was continuously produced for over 3,000 hours at an average rate of 50-100 kJ/hr [14 to 28 watts]" Arata also described a model that he believes explains the reaction, and accounts for the good performance of palladium black in pure deuterium gas.

A number of Japanese corporations showed up with mainstream CF results that would have described as "spectacular" a few years ago, including large heat bursts, boil-offs, and the like. Iwamura [11], from Mitsubishi Heavy Industries, reported X-rays, neutron emissions and possible transmutations, and concluded, "Although we cannot identify where these Pd atoms came from (contamination or generation), we can say that anomalous nuclear reactions must occur in the electrochemical cells at room temperature." Itoh [12], also from Mitsubishi, reported on vacuum chamber gas release experiments somewhat similar to the NTT thin film work reported at ICCF3 and elsewhere. Shikano [13] of NTT reported continuing progress with those experiments.

Isagawa [14], with the Japanese National Laboratory for High Energy Physics (KEK), got a number of spectacular results, including three boiling events and an "enormous" heat burst. "Under constant current conditions, the cell voltage and the cell temperature were increased gradually and all of a sudden sharply increased to boiling. . It was just during the calm period about 6 hours after the first boiling that the enormous heat release was observed. The temperature of the cell of about 100 ml in volume increase by 7.5 K (from 83.4 deg C to 90.9 deg C) in 13 minutes. The cell voltage showed a dip correspondingly. The excess heat can be estimated to be 6.8 W, about 110% with respect to the input electrical power. . . . Boiling occurred 3 times, the last episode continuing for about 16 hours, in the former period violent but in the prolonged later period rather gentle; the cell was driven almost to dryness." KEK has superb instrumentation so nobody can deny the results are real and beyond chemistry.

Claytor's abstract [15] reports continued progress at Los Alamos. "we have been able to demonstrate that a plasma loading method produces an exciting and unexpected amount of tritium. In contrast to electrochemical [methods], this method yields a reproducible tritium generation rate".

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COLD FUSION TIMES

The Journal Of The Scientific Aspects Of Loading Isotopic Fuels Into Materials

PAGE REPORT ON ICCF-5

[The following excerpt from runner-up Bill Page]

"An apparently 300% excess heat demonstration was operated by Dennis Cravens and Clean Energy Technologies, Inc. (CETI). The experimental apparatus, which was completely enclosed in a plastic display case with panels opening to allow access to the electrolytic cell and measurement equipment. The demonstration cell used light water (though heavy water has also been tested) and a cathode consisting of thin Nickel/Palladium/Nickel/Copper layers on several thousand small plastic polymer beads. The beads are held in close vicinity to a resin ion exchange anode. The electrolyte is pumped at a constant rate through the cell. Flow calorimetry on the circulating electrolyte is used to measure the heat evolved during the electrolysis. CETI encouraged conference participants to make their own measurements on the operating cell and many people seemed to attempt to do so as everyone crowded around the display during coffee breaks and even during the other conference sessions. McKubre observes large and frequent absorption/desorption cycles after achieving loadings of above about 0.83 (which Martin Fleischmann and others also noted corresponds to the endothermic to exothermic absorption transition) coincident with the observation of excess heat. These absorption/desorption oscillations were also noted by another speaker. Peter Hagelstein has a theory that desorption can be the mechanism which pumps the phonon-laser effects which are required by his theory in order to account for the lack of radiation products from fusion .. based on neutron transfer and phonon laser effects. Peter Hagelstein's theory has the merit of having survived (and evolved) as "CF" research has continued. There were 31 other theory papers presented as posters, of which mine was one. "

§ §

COLD FUSION TIMES

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TO COLDLY GO WHERE NO ONE HAS GONE

MORE CF REPORTS:

From China, India, Italy, France, Massachusetts, Monaco, Romania, Russia and Utah and augmenting the copious ICCF-5 news inside, much more has been going on.

Italian scientists have continued calorimetric and material exploration in this area.

BOLOGNA, Italy (Reuter) - Italian physicists have detected what they believe may be evidence of nuclear ``cold fusion" in experiments with hydrogen and nickel, the news agency AGI reported. It quoted a member of the team, Professor Sergio Focardi of the University of Bologna, as saying that 15 grams of nickel and one gram of hydrogen produced 30 to 40 watts of energy, sufficient to power a light bulb, for around three months.

AGI quoted Focardi as saying: **``We have the certainty that the system we are experimenting with produces energy through a process that is perfectly replicable and controllable. Measures based on controlling the external temperature of the system prove without a shadow of doubt that energy is emitted from this system."**

Scientists have long theorized that fusion, the process that powers the sun, could provide all the world's electricity needs without using any fossil fuel if harnessed. Many physicists have dismissed cold fusion -- creating the type of energy used by the sun without extremely high temperatures -- as a delusion. Focardi said about 100 kilowatt hours of energy had been produced so far in the experiments, a quantity that ruled out a chemical reaction as the source of the power. **``We have observed neutrons and gamma rays, which are fortunately few and can easily be eliminated. The presence of this radiation is proof that nuclear reactions are occurring but they cannot be explained with current knowledge of physics.'** §§

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P.O. Box 81135
Wellesley Hills, MA USA 02181

TO: